

Global Issues in Engineering Ethics

Introduction to Global Ethics

- Globalization integrates nations via trade, technology, and culture.
- Engineers and MNCs play key roles in globalization.
- Ethical considerations include business, environment, and technology.

Multinational Companies (MNCs)

- MNCs operate in home and host countries.
- Influence local and global economies.
- Responsibilities: Wages, safety, employee benefits, ethical operations.

Business Ethics

- Set of moral guidelines for business conduct.
- Includes employee welfare, environmental care, waste management.
- Promotes trust, transparency, and social good.

Environmental Ethics

- Focus on moral issues affecting the environment.
- Pollution, resource depletion, global warming are major concerns.
- Engineers must develop eco-friendly solutions and sustainable designs.

Computer Ethics

- Concerns with privacy, property, consent, and social harm.
- Ten Commandments of Computer Ethics include avoiding harm, respecting property, and ensuring honesty.
- Growth of cybercrime is an ethical challenge.

Computers in Technological Development

- Computers enhance communication and commerce.
- Issues: Free speech, financial fraud, software theft, identity theft.
- Geographical coverage makes tracking cybercrimes difficult.

Privacy Factors

- Cyber threats risk personal and corporate data.
- Viruses, hacking, data breaches compromise security.
- Legal protections like the Privacy Act of 1947 exist.

Weapons Development

- Engineering contributes to modern weapons.
- Ethical dilemma: National defense vs. human harm.
- Engineers may feel conflicted about military applications.

Engineers as Managers

- Engineers must resolve conflicts and promote ethical policies.
- Balance between profitability and social responsibility.
- Ensure safe and valuable product development.

Managing Conflicts

- Conflicts arise from scheduling, priorities, resources, and personalities.
- Managers must resolve conflicts with patience and ethical focus.
- Objective: Efficiency, quality, customer satisfaction.

Consulting Engineers

- Provide specialized knowledge independently.
- Maintain ethical clarity in consulting and avoid conflicts of interest.
- Offer unbiased expert advice.

Engineers as Advisors

- Must recognize limitations and work with qualified experts.
- Adopt roles: Hired Guns, Value-Neutral Analysts, Value-Guided Advocates.
- Honesty and public welfare must guide decisions.

Conclusion

- Global ethics demand responsibility in engineering.
- Balancing innovation, profit, and social good is key.
- Continuous ethical practice ensures long-term success.

GLOBAL ISSUES IN ENGINEERING ETHICS

CASE STUDIES: GRAMEENPHONE & UNILEVER



INTRODUCTION TO GLOBAL ETHICS

- Ethics in engineering has global implications.
- Multinational companies must balance profit with social responsibility.
- Case focus: Grameenphone & Unilever in Bangladesh.

WHAT IS ENGINEERING ETHICS?

- Commitment to public welfare, honesty, and safety.
- Applies to business decisions, technology, and societal impact.

GLOBALIZATION AND ETHICS

- Integration of economies and cultures.
- Companies operate across diverse legal and ethical systems.
- Challenges: Fairness, sustainability, cultural respect.

ROLE OF MULTINATIONAL COMPANIES

- MNCs like Grameenphone and Unilever shape global business.
- Influence on local economies, labor practices, and environment.
- Need for ethical frameworks to guide operations.

BUSINESS ETHICS OVERVIEW

- Ethical treatment of employees and stakeholders.
- Transparent accounting and fair practices.
- Social good and environmental care.

GRAMEENPHONE: BUSINESS ETHICS

- Largest telecom operator in Bangladesh.
- Digital inclusion vs. data privacy concerns.
- Fair pricing and customer service ethics.

UNILEVER: BUSINESS ETHICS

- Consumer goods giant with sustainable living goals.
- Challenges: Plastic waste, ethical sourcing.
- Community development programs in Bangladesh.

ENVIRONMENTAL ETHICS

- Industrial growth impacts the environment.
- Both Grameenphone and Unilever face sustainability challenges.
- Need for eco-friendly operations.

GRAMEENPHONE: ENVIRONMENTAL EFFORTS

- Network energy efficiency programs.
- E-waste recycling initiatives.
- Carbon footprint management.

UNILEVER: ENVIRONMENTAL COMMITMENTS

- Plastic reduction initiatives.
- Water conservation in production.
- Promoting reusable packaging.



COMPUTER AND DATA ETHICS

- Digital businesses must protect privacy.
- Ethical use of customer data is critical.
- Transparency in AI and algorithms.

GRAMEENPHONE: DATA ETHICS

- Managing customer data securely.
- Challenges in digital rights and surveillance concerns.
- Transparent policies for data usage.

CORPORATE SOCIAL RESPONSIBILITY (CSR)

- CSR links ethics to business strategy.
- Enhances reputation and builds trust.
- Both companies run social programs in Bangladesh.

UNILEVER: SOCIAL INITIATIVES

- Empowering women through small businesses.
- Health and hygiene education programs.
- Supporting local agriculture.

MANAGING ETHICAL CONFLICTS

- Balancing profit with social good is challenging.
- Conflicts in pricing, labor rights, and environmental impact.
- Need for ethical leadership and policies.

REGULATIONS AND STANDARDS

- Global and local laws guide ethical behavior.
- Compliance with labor, environmental, and data regulations.
- Voluntary codes of conduct support ethical business.

ETHICAL LEADERSHIP

- Leaders set the tone for corporate ethics.
- Example: Ethical CEO decisions influence company culture.
- Transparent communication is key.

FUTURE OF GLOBAL ETHICS IN BUSINESS

- Focus on sustainable development.
- Responsible innovation and technology use.
- Collaborating with stakeholders for long-term impact.



LESSONS LEARNED FROM GRAMEENPHONE & UNILEVER

- Integrate ethics into business models.
- Continuous improvement in sustainability.
- Respect for cultural, social, and environmental contexts.



CONCLUSION

- Global ethics is vital for modern engineering and business.
- Grameenphone and Unilever provide case studies of ethical practice.
- Building a responsible and sustainable future is a shared duty.

GLOBAL ISSUES IN ENGINEERING ETHICS

DATA PRIVACY CONCERNS IN HUAWEI'S OPERATIONS



INTRODUCTION

- Globalization brings ethical challenges for tech companies.
- Huawei's data privacy practices raise global concerns.
- Focus: Engineering ethics and data privacy in Huawei.

WHAT IS ENGINEERING ETHICS?

- Protect public welfare, safety, and privacy.
- Ensure honesty, accountability, and transparency in technology.

DATA PRIVACY:A GLOBAL ISSUE

- Personal data is valuable and vulnerable.
- Engineers must protect data integrity and security.
- Data misuse leads to trust erosion and legal risks.

HUAWEI OVERVIEW

- Leading global telecom and tech company.
- Operations in over 170 countries.
- Criticized for alleged data privacy violations.

ETHICAL CONCERNS WITH HUAWEI

- Accusations of unauthorized data collection.
- Potential ties to state surveillance.
- Global debate over security and privacy.

ENGINEERING RESPONSIBILITIES

- Design secure systems protecting user data.
- Implement ethical AI and IoT data practices.
- Avoid hidden data flows and backdoors.

GLOBAL DATA PRIVACY STANDARDS

- GDPR (Europe): User consent, data minimization.
- CCPA (California): Consumer data rights.
- Huawei must align with international regulations.

HUAWEI AND GDPR COMPLIANCE

- Huawei claims compliance with GDPR.
- Concerns remain over data storage and transfer.
- Need for transparent data practices.

SECURITY RISKS IN 5G NETWORKS

- Huawei is a global 5G leader.
- Critics fear potential for espionage via infrastructure.
- Calls for independent audits and verification.

ENGINEERING ETHICS: DATA PROTECTION

- Confidentiality: Keep user data safe.
- Integrity: Prevent data tampering.
- Availability: Ensure data is accessible to rightful owners only.

HUAWEI'S DATA HANDLING PRACTICES

- End-to-end encryption in some services.
- Data localization efforts in compliance regions.
- Questions about backend data access remain.

TRANSPARENCY AND TRUST

- Transparency is key to public trust.
- Huawei's lack of open source scrutiny fuels skepticism.
- Need for verifiable ethical commitments.

WHISTLEBLOWING AND ETHICAL DILEMMAS

- Engineers face dilemmas if asked to compromise privacy.
- Whistleblowing can prevent harm but risks personal loss.
- Ethical culture is crucial.



ROLE OF GOVERNMENT AND REGULATION

- Governments must enforce privacy regulations.
- International cooperation needed for global companies.
- Ethical auditing as a regulatory requirement.

CORPORATE SOCIAL RESPONSIBILITY (CSR)

- CSR includes protecting digital rights.
- Huawei invests in social programs but faces ethical scrutiny.
- Balancing profit with ethical tech development.

ETHICAL DESIGN AND DEVELOPMENT

- Integrate privacy by design in all engineering phases.
- Continuous ethical training for engineers.
- Adopt global ethical guidelines for data handling.



CASE STUDIES AND CONTROVERSIES

- Huawei's ban in the US and some EU countries.
- Allegations of backdoor access in telecom equipment.
- Response: Huawei denies involvement in unethical practices.

LESSONS LEARNED

- Data privacy must be prioritized in engineering decisions.
- Global companies must adhere to ethical standards everywhere.
- Transparency and independent audits build trust.

FUTURE OF DATA PRIVACY IN TECH

- Growing need for data ethics in AI, IoT, and telecom.
- Privacy-centric innovation is essential.
- Global consensus on data ethics is emerging.

CONCLUSION

- Huawei's case highlights global ethical challenges in tech.
- Engineers must champion data privacy and ethical practices.
- Building trust requires ongoing commitment to ethics.